

VOLUME 3 • B.TECH CSE STUDY SERIES

JAVA PROGRAMMING

Complete Professional Handwritten Notes

JDK, JRE & JVM Internals, Full OOP Paradigm, Collections Framework, Multithreading, Streams & Enterprise Prep

```
public class Main {  
    public static void ma  
        System.out.println(  
}
```

✓ JVM Architecture & Memory Model

✓ OOP Principles (Interfaces & Abstract)

✓ Java Collections Framework

✓ Multithreading & Synchronization

✓ Java 8+ Streams & Lambdas

✓ 30+ Viva & Enterprise Interview Q&A

Designed for B.Tech Exams, Campus Placements & Enterprise Backend Roles



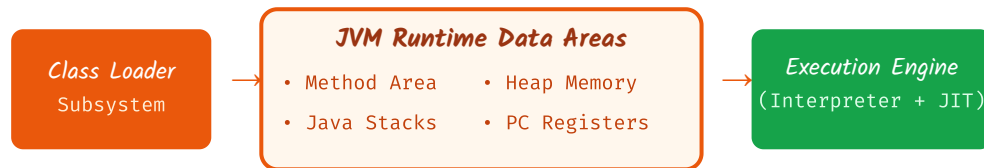
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CHAPTER 1: JVM ARCHITECTURE & COMPILATION FLOW

💡 JDK vs JRE vs JVM Architecture

SVG DIAGRAM: JVM MEMORY & SUBSYSTEM ARCHITECTURE



2. Code Example: Platform Independence

```
public class JVMTest {  
    public static void main(String[] args) {  
        System.out.println("Write Once, Run Anywhere (WORA)");  
    }  
}
```

CHAPTER 8: JAVA COLLECTIONS FRAMEWORK

Collections Hierarchy Overview

Interface	Implementation	Ordered?	Duplicate Allowed?	Key Feature
List	ArrayList	Yes	Yes	Fast random access (index-based). Dynamic array resize.
List	LinkedList	Yes	Yes	Doubly-linked list. Fast insertions/deletions.
Set	HashSet	No	No	Backed by HashMap. $O(1)$ average search.
Set	TreeSet	Yes (Sorted)	No	Red-Black tree implementation. Elements sorted.
Map	HashMap	No	Keys No, Values Yes	Key-value pairs. $O(1)$ performance.

```
import java.util.*;

public class CollectionDemo {
    public static void main(String[] args) {
        List<String> list = new ArrayList<>();
        list.add("Java");
        list.add("Python");
        list.add("C++");

        list.forEach(item → System.out.println("Tech: " + item));
    }
}
```

CHAPTER 14: 30+ UNIVERSITY VIVA & INTERVIEW QUESTIONS

Q1. Why is String immutable in Java?

Ans: Security, String Pooling (saving Heap memory), Thread Safety, and Caching hashCode for fast HashMap key lookup.

Q2. What is the difference between final, finally, and finalize?

Ans: final is a keyword to restrict class/method/variable inheritance or modification. finally is a block used with try-catch for cleanup. finalize() is a method called before Garbage Collection.